

13. Potential Risks and Mitigation Strategies

Table 9. Risks and Mitigation

Risk	Category	Impact	Mitigation Strategy
Substandard or Unusable Donated Items: Donors submit damaged textbooks, worn-out uniforms, or obsolete/heavily damaged IT hardware beyond repair, leading to wasted transport and storage overhead.	Operational / Business	High	•Require donors to upload authentic item photos and fill out a self-declaration condition checklist on the Donor Portal. •Enforce a standard 3-tier physical verification process (Grade A / Grade B / Grade C) immediately at the intake phase. •Establish formal partnerships with certified e-waste recycling facilities to handle irreparable electronic scrap responsibly.
Data Disputes & Multi-Warehouse Inventory Conflicts (Race Conditions): Multiple coordinators simultaneously allocate the same shipment during a campaign peak ("Relay to School") leading to out of stock or deviation from actual quantity.	Technique	Cao	•Apply distributed locking mechanism (Distributed Locks) to hold inventory instantly. •Strictly enforce ACID integrity transactions with Optimistic Locking on PostgreSQL. •Record immutable audit logs for each step in changing the status of an artifact (Pending approval → Inspected → Ready to distribute → Released).
Leakage of privacy and Disadvantaged Student Data: Publicizing family information, images or identities of disadvantaged students violates regulations on personal data protection and children's rights.	Bảo mật / Pháp lý	Cao	•Perform Data Masking & Pseudonymization for all student data on public lookup portals. •Limit access to detailed records to authorized school representatives and administrators only. •Apply RBAC decentralization mechanism and encrypt all data transmitted via TLS 1.3 protocol.

Risk	Category	Impact	Mitigation Strategy
Fraud or exaggeration of support needs support: Unverified organizations/individuals send fake lists of students to appropriate high-value devices (laptops, desktops, tablets).	Quản trị / Kiểm soát	Cao	•Required legal authentication of the school: required to upload a written request signed by the Board of Directors and stamped in red. •Implement the digital handover process (Proof of Delivery - PoD): take photos at the scene with GPS positioning, scan the QR code and sign electronically.
Volunteers absent or canceling shifts at the last minute: Volunteers registered but did not show up to do sorting duties at the warehouse or deliver goods during the peak period at the beginning of the school year.	Personnel operations	Average	•Build a list of backup volunteers (schedule 15-20% of the army). •Send automatic calendar reminders via Push Notification (Firebase FCM) and Email 24 hours before shift. •System integration Accumulate hours of dedication and issue an electronic Volunteer Certificate.
Supply and demand mismatch between geographical regions: There is too much donated goods in big cities (like Da Nang) while schools in remote areas (like the mountainous areas of Quang Nam) have a serious shortage.	Logistics / Coordination	Average	•Historically intelligent dispatch algorithm in Python to automatically propose internal transfer orders between nearby warehouses. •Update the shortage list in real time to the campaign homepage to direct the community to donate exactly what is needed.

Risk	Category	Impact	Mitigation Strategy
Interruption of network connection and Third Party Services: 3G/4G signal flickers in highland communes when making deliveries, or cloud storage service/notification of technical problems.	Technique	Average	•Design a React Native mobile application according to Offline-first architecture: store temporary data locally and automatically synchronize when there is a network connection. •Configure asynchronous processing queue (Task Queue) with automatic retry mechanism (Retry) when calling external API.
Project Completion progress multi-party business process (Receiving, Inspection, Repair, Warehouse Coordination, Volunteers, ESG Report) exceeding the 14–16 week time frame of the project.	Time	Average	•Strictly follow the Agile Scrum process with a 3-week/Sprint development cycle. •Prioritize completion of the core MVP modules (Donations, QR Code Assignment, Warehouse Management, Request Receiving, Coordination, PoD Handover, Dashboard).

14. Expected Outcomes / Deliverables

- **EduShare Vietnam Web/Application System:** Web and mobile application system Completely allows for comprehensive digitization of the process of receiving educational resources from home sponsorship, multi-point warehouse management with QR identification code, quality inspection, refurbishment computer repair, coordination of automatic allocation to disadvantaged schools and according to Transparently track the handover journey in real time.
- **Source Code:** Complete source code of the application (front-end and back-end) and system installation and configuration instructions.

- **Technical Documentation:**

- Proposal Document
- Project Plan Document
- Product Backlog Document
- User Stories Document
- Architecture Document
- Database Design
- UI Design
- Test Plan Document
- Test Case
- Sprint Backlog
- Code Standard
- Meeting Document
- Reflection Document

- **Final Report:**

- Comprehensive reports on the entire system research and development process.
- Analyze achieved results compared to initial goals.
- Evaluate limitations and suggest future directions for improvement, e.g
Such as AI applications that classify books and equipment through images, and AI that estimates usage value remaining, integrating Blockchain technology to make transaction logs transparent, and social impact analysis (ESG) dashboard for sponsoring businesses.

15. References

Table 10. References

No.	References	Document Information
1	Scrum Model	https://www.atlassian.com/agile/scrum
		https://www.digite.com/agile/scrum-methodology/
		https://docs.microsoft.com/en-us/azure/devops/boards/sprints/best-practices-scrum?view=azure-devops
		https://www.scrum.org/resources/scrum-guide
2	Technical	https://nodejs.org/en/learn/getting-started/introduction-to-nodejs
		https://aws.amazon.com/vi/what-is/api/
		https://microservices.io/
3.	Software Engineering Standards	https://www.nws.noaa.gov/oh/hrl/developers_docs/General_Software_Standards.pdf
		https://standards.ieee.org/standard/12208-2017.html
		https://sw-eng.larc.nasa.gov/

DESCRIPTION OF PRODUCT REQUIREMENTS"**16.1. Brief description of product ideas (less than 7 statements).****16.2. Requirements**

High-level functional requirements	1. Emergency Rescue Request Submission (SOS)
	2. GPS-based Location Detection and Map Display
	3. Rescue Provider / Volunteer Assignment
	4. Real-time Rescue Status Tracking
	5. User and Rescuer Account Management
	6. Notification and Alert System
	7. Rescue History and Reporting

Quality property requirements (e.g., related to issues: Easy to Use, Easy to Like, Easy to Learn, Easy to Understand, Easy to Buy / Yes,)	1. Easy to use
	2. Easy to learn
	3. Easy to interact
	4. Easy to understand
	5. Easy to monitor and analyze rescue statistics

Operational requirements (related to the following issues: Speed, Accuracy, Performance, Stability, Load Protection, Scalability, Safety, ...)	1. Ensure system stability and reliability
	2. Ensure accuracy of location and rescue information
	3. Ensure system security and data protection
	4. Support high concurrent access during emergency situations
	5. Ensure system scalability for future expansion

Environmental and operational requirements (related to the following issues: physical impact on the environment, interaction with relevant or existing systems, conditions of commercialization of products, ...)	1. Can be accessed on multiple web platforms (desktop, tablet, mobile browsers)
	2. Can integrate with external services such as map APIs and notification services
	3. Operates effectively under real-time conditions

Maintenance and support requirements	1. Easily maintain source code
	2. Easily expand source code
	3. Specific description of the document

Security/safety requirements (related to the following issues: conditions of use / access to products, personal freedom, inspection, ...)	1. User information and rescue data must be securely protected
	2. Secure authentication and authorization mechanisms
	3. Anti-intrusion and attack prevention mechanisms

Assess the complexity of technical problems		1. Regarding broad technical issues or contradictions
		2. No clear solution
		3. Resolve issues that are not covered by current standards and codes
	X	4. Relate to different stakeholder groups
	X	5. Include multiple component parts or sub-issues
	X	6. Related to many disciplines
	X	7. Have significant consequences in a wide range of context.

Standard requirements	X	1. Code standard.
	X	2. Design standards. (object-oriented design, analysis, and design templates...).
		3. IEEE (1058, 1540, 830, 1016, 829, 1012, 1008)
	X	4. ISO/IEC/IEEE 12207:2017 (TCVN 10539:2014); ISO/IEC 25051:2006 (TCVN 10540:2014);
		5. Other standards. (related to specific topics)